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Peptide Reconstitution Guide

A practical reference for in-vitro research

Dosage Calculator | Cycle Calculator | Price Comparison

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What Is Reconstitution?

Peptides are supplied as lyophilised (freeze-dried) powder. Reconstitution is the process of adding bacteriostatic water (BAC water) to turn that powder into a liquid solution that can be measured and dosed with an insulin syringe.

Supplies You'll Need

Item	Purpose
Peptide vial	Lyophilised powder containing the peptide
Bacteriostatic water	Sterile diluent with 0.9% benzyl alcohol preservative
1 mL insulin syringe	Measuring and transferring liquid
Alcohol swabs	Sanitising vial tops before puncture
Sharps container	Safe disposal of used syringes
Gloves	Sterile handling

Reconstitution Protocol

Before You Start

Bring peptide and BAC water to room temperature (approx. 30 minutes) to prevent condensation and cloudiness. Wash hands and put on gloves. Sanitise work surface and prepare all supplies.

1. **Sanitise**
   Clean both vial stoppers (peptide and BAC water) thoroughly with an alcohol swab.
2. **Draw BAC Water**
   Open the insulin syringe. Pull back the plunger to your desired BAC water volume (commonly 1-2 mL). Insert the needle into the BAC water vial at 90 degrees. Inject air to equalise pressure, then draw the water.
3. **Transfer to Peptide Vial**
   Insert the needle into the peptide vial at a 45-degree angle with the bevel facing up. Slowly push the water down the inside wall of the vial. Do not spray directly onto the powder. Remove syringe and dispose safely.
4. **Dissolve**
   Do not shake. Gently swirl or roll the vial until the powder fully dissolves. If powder remains after 1-2 minutes, let sit for 5-10 minutes then swirl again. The solution should be clear with no visible particles.
5. **Store**
   Label the vial with the reconstitution date and concentration. Refrigerate immediately at 2-8 degrees C. Do not freeze after reconstitution.

Dosage Quick Reference

The calculation is based on three inputs: vial strength, water added, and target dose.

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Concentration (mcg/mL) = (vial mg x 1000) / water added (mL)
mcg per unit = Concentration / 100
Units to draw = target dose (mcg) / mcg per unit
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Common Ratios

Vial	BAC Water	Per Unit	250mcg	500mcg	1mg
5 mg	1 mL	50 mcg	5 u	10 u	20 u
5 mg	2 mL	25 mcg	10 u	20 u	40 u
10 mg	2 mL	50 mcg	5 u	10 u	20 u
10 mg	3 mL	33.3 mcg	7.5 u	15 u	30 u
15 mg	1 mL	150 mcg	1.7 u	3.3 u	6.7 u
15 mg	3 mL	50 mcg	5 u	10 u	20 u
20 mg	2 mL	100 mcg	2.5 u	5 u	10 u
20 mg	4 mL	50 mcg	5 u	10 u	20 u

Syringe Reference

Syringe	Total Units	Best For
1.0 mL	100 units	Most doses (standard choice)
0.5 mL	50 units	Small doses; markings are wider apart
0.3 mL	30 units	Very small doses; most precise

All standard insulin syringes use the U-100 scale. The units are identical regardless of barrel size; a smaller barrel simply spreads the markings further apart for easier reading.

Storage Guidelines

State	Temperature	Shelf Life
Lyophilised (long-term)	-20 degrees C or below	1 year+
Lyophilised (short-term)	2-8 degrees C	1-3 months
Lyophilised (room temp)	~20 degrees C (dark, dry)	2-3 weeks
Reconstituted solution	2-8 degrees C	Up to 4 weeks
Aliquoted (single-dose frozen)	-20 degrees C	3-4 months

Key Rules

- **Keep away from light.** Store in original box or opaque container.
- **Do not freeze after reconstitution.** Unless aliquoted into single-use portions. Freeze-thaw cycles degrade peptides.
- **Discard if:** Solution is cloudy, contains particles, shows discolouration, or exceeds 4 weeks post-reconstitution.
- **BAC water:** Once opened, refrigerate and use within 4 weeks.

Common Mistakes to Avoid

- **Shaking the vial** -- Creates foam and can denature the peptide. Always swirl or roll gently.
- **Blasting water onto powder** -- Damages peptide structure. Inject slowly down the vial wall at 45 degrees.
- **Using plain sterile water** -- No preservative; bacteria can grow within 24 hours. Always use BAC water (0.9% benzyl alcohol) for multi-dose vials.
- **Reusing syringes** -- Contamination risk. Use a fresh sterile syringe for each draw.
- **Storing at room temperature** -- Accelerates degradation. Refrigerate immediately at 2-8 degrees C.
- **Freezing reconstituted solution** -- Ice crystals damage peptide chains. Keep refrigerated, not frozen.

Frequently Asked Questions

Can I use sterile water instead of BAC water?

Sterile water has no preservative -- once punctured, bacteria can grow within 24 hours. BAC water contains 0.9% benzyl alcohol which inhibits bacterial growth for multiple uses over approximately 28 days. Always use BAC water for multi-dose vials.

How long is a reconstituted peptide good for?

Up to 4 weeks when refrigerated at 2-8 degrees C. Some peptides may degrade sooner depending on stability. Always check the manufacturer's guidelines and discard if the solution appears cloudy or discoloured.

Why is my solution cloudy or foamy?

Cloudiness can indicate contamination, peptide degradation, or improper reconstitution. Foaming usually results from shaking or blasting water directly onto the powder. If cloudy, do not use. If foamy, let it settle before use.

Can I mix two different peptides in one syringe?

Most peptides should not be mixed without thorough research. Mixing can cause precipitation, degradation, or unexpected chemical reactions. Check compatibility data for your specific peptide combination before attempting.

How do I measure very small doses accurately?

Use a 0.3 mL (30 unit) or 0.5 mL (50 unit) insulin syringe. The smaller barrel spreads the markings further apart for better precision. You can also adjust your water volume to create a less concentrated solution.

My peptide powder won't fully dissolve. What should I do?

Some peptides take longer to dissolve than others. Let the vial sit for 5-10 minutes, then gently swirl again. Avoid heating or shaking. If it still won't dissolve, the peptide may have been exposed to moisture or heat during storage.
